Nixor O Levels
Test Week 1
Mathematics
Grade 10

Student Name

Student ID:

Section :

Time: 60 minutes

Max marks: 40

Please read and follow the instructions below:

- Please write your name, section and ID on the question paper.
- Do not turn over the paper unless instructed to do so.
- Only write in blue or black ink.
- Use of correction fluid is strictly prohibited.
- If you have any questions, raise your hand to ask the invigilator.
- Omission of working will result in loss of marks.
- Use a pencil for making stralght lines.

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$\frac{\sin a}{a} = \frac{\sin b}{b}$

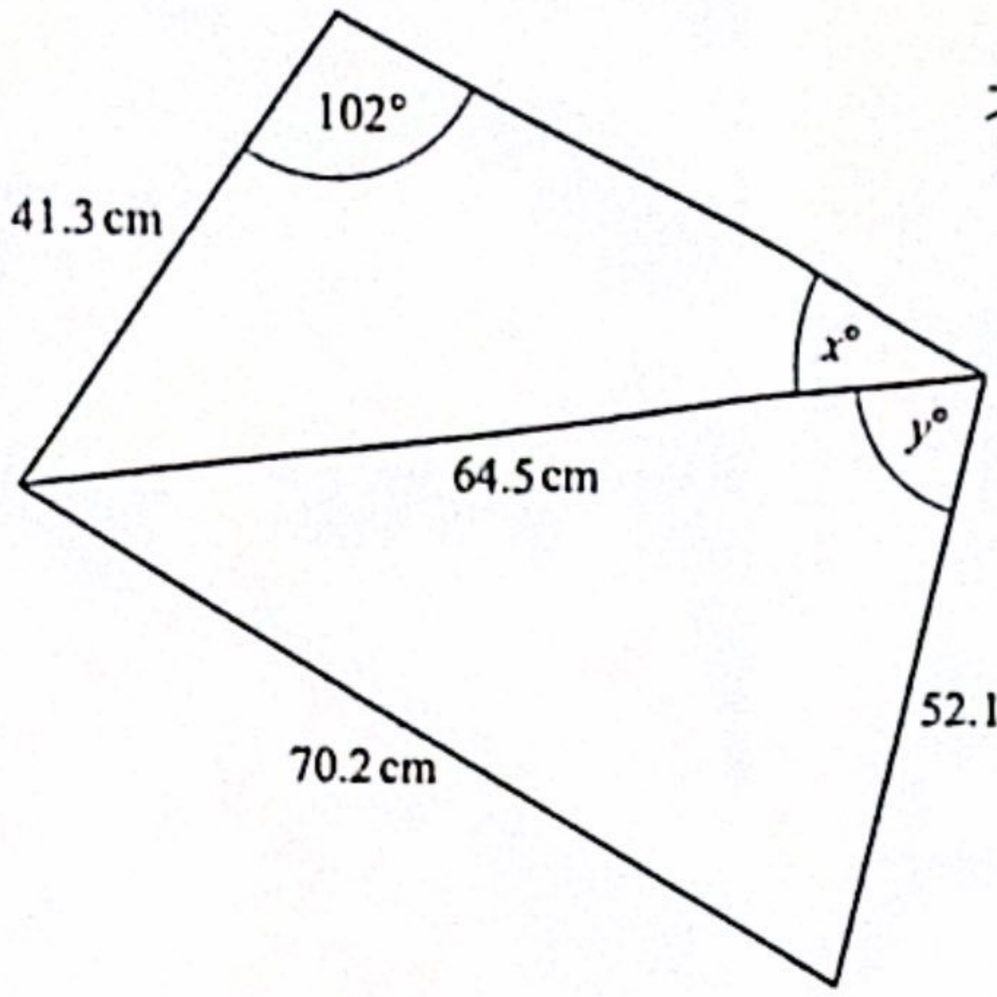
$a^2 = b^2 + c^2 - 2bc\cos A$

$\frac{1}{2} a \times b \sin C$

39

q1.

PCL



NOT TO SCALE

(a) Calculate the value of x.

$\frac{\sin x}{41.3} = \frac{\sin 102}{64.5}$

$\sin x = \frac{41.3 \sin 102}{64.5}$

$\sin x = 0.62631$

$x = \sin^{-1}(0.62631)$

$x = 38.7789$

$\approx 38.8°$

$x =$ 38.8° [3]

PCC

(b) Calculate the value of y.

$\cos y = \frac{64.5^2 + 52.1^2 - 70.2^2}{2 \times 64.5 \times 52.1}$

$\cos y = 0.289636$

$y = \cos^{-1}(0.289636)$

$y = 73.16378$

$\approx 73.2°$

$y =$ 73.2° [3]

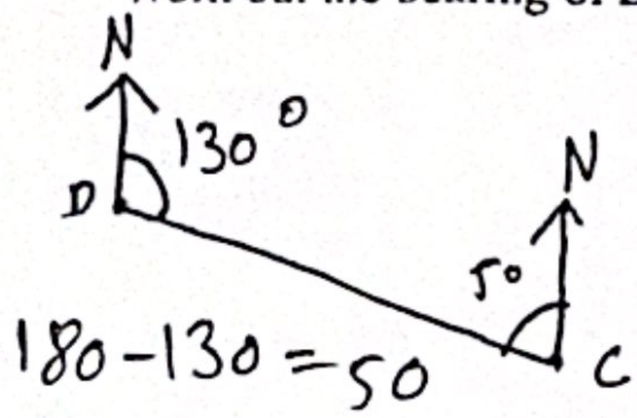
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PCL

Q2.

The bearing of C from D is 130°.

Work out the bearing of D from C.



180 − 130 = 50

360 − 50 = 310°

310° [2]

VCL Q3.

Martha walks a distance of 10 km at a speed of x km/h.
She then runs a distance of 5 km at a speed of $(x+4)$ km/h.
The total time taken for the whole journey is 3.5 hours.

(a) Write down an expression in terms of x for the time Martha is walking.

$S = \frac{D}{t} \rightarrow t = \frac{D}{S} = \frac{10\text{ km}}{x\text{ km/h}}$

$\frac{10}{x}$ h [1]

VCL (b) Show that $7x^2 - 2x - 80 = 0$.

$$\frac{10}{x} + \frac{5}{x+4} = 3.5$$

$$\frac{10x + 40 + 5x}{x^2 + 4x} = 3.5$$

$$10x + 15x + 40 = 3.5x^2 + 14x$$

$$0 = 3.5x^2 + 14x - 15x - 40$$

$(0 = 3.5x^2 - x - 40) \times 2$

$$0 = 7x^2 - 2x - 80$$

[4]

(c) Solve $7x^2 - 2x - 80 = 0$, giving your answers correct to 2 decimal places.
You must show all your working.

$$\frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 7 \times -80}}{2 \times 7} \rightarrow \frac{2 \pm \sqrt{4 + 2240}}{14} \rightarrow \frac{2 \pm \approx 47.37}{14}$$

1) $\frac{2 + 47.37}{14} \rightarrow \frac{49.37}{14} = 3.5264$
$\times \approx 3.53$

2) $\frac{2 - 47.37}{14} \Rightarrow \frac{-45.37}{14} \Rightarrow -3.241$
$\times \approx -3.24$

$x =$ 3.53 or $x =$ -3.24 [3]

PLL

(d) Calculate the difference between the time Martha is walking and the time she is running.
Give your answer in hours and minutes correct to the nearest minute.

$\frac{10}{3.53} - \frac{5}{3.53+4}$

$\frac{10}{3.53} - \frac{5}{7.53}$

$2.832 - 0.664$

$= 2.168$

≈ 2.17

2 hours ~~2~~ 10 mins

$2.17 - 2 = 0.17 \times 60 = 10.2$

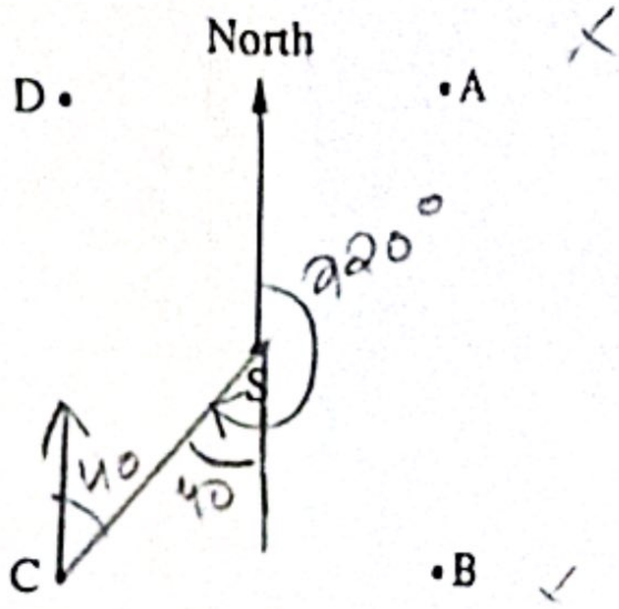
$10.2 \approx 10$

2 hours 10 mins

..... h ..... min [3]

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Q4.



The bearing of a lighthouse from a ship, S, is 220°.
The position of S is marked on the diagram.

(a) Which of the four points A, B, C or D is a possible position of the lighthouse?

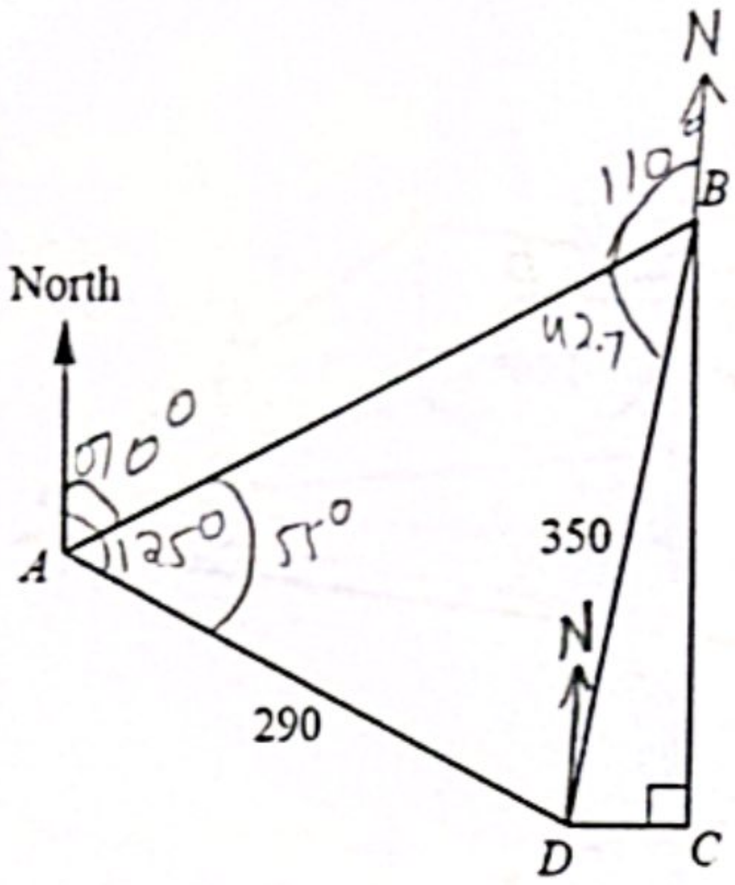
Answer (*a*) C [1]

(b) Write down the bearing of S from the lighthouse.

220-180 = 40°

Answer (*b*) 040° [1]

Q5.



NOT TO SCALE

ABCD is a field on horizontal ground.

The bearing of *B* from *A* is 070°.
The bearing of *D* from *A* is 125°.
C is due south of *B* and due east of *D*.
AD = 290 m and *BD* = 350 m.

(a) Calculate the bearing of *D* from *B*.

$\frac{\sin A\hat{B}D}{290} = \frac{\sin 57}{350}$

$\sin A\hat{B}D = \frac{290 \sin 57}{350}$

$A\hat{B}D = \sin^{-1}\left(\frac{290 \sin 57}{350}\right)$

$A\hat{B}D = 42.744$

~~180 − 110 − 42.744 =~~

360 − 110 − 42.744 = 207.255

~~207~~

≈ 207°

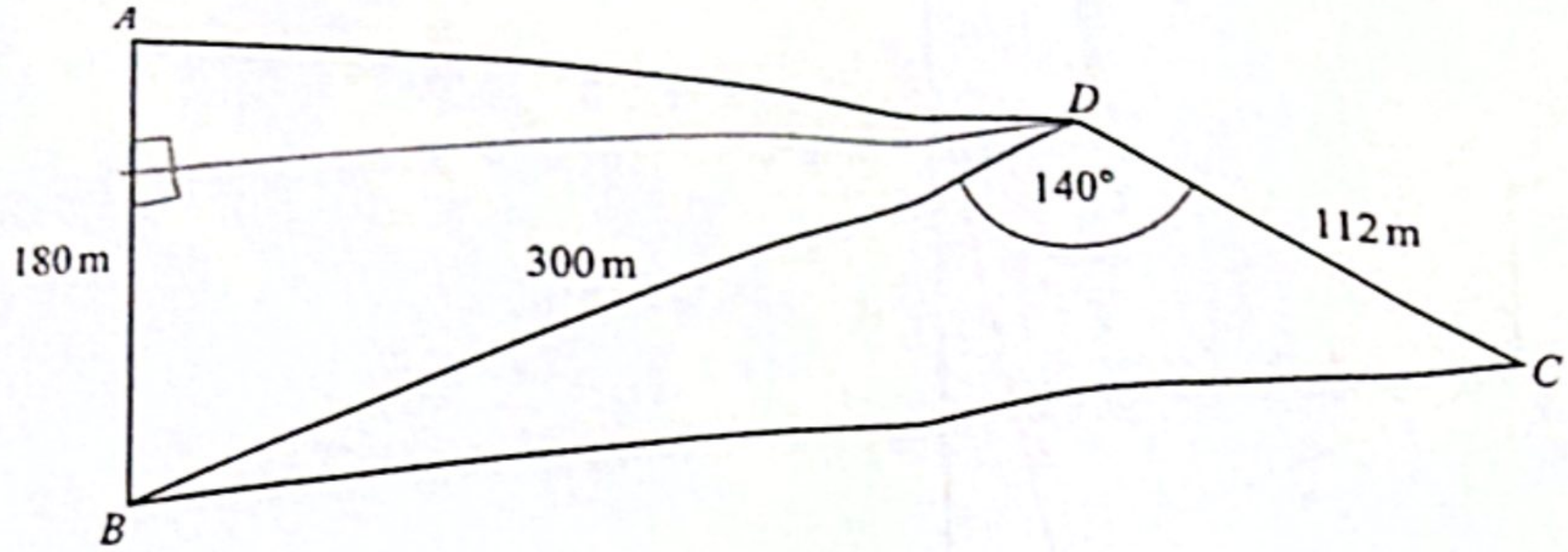
207°

..... [4]

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OCU

Q6.



NOT TO SCALE

The diagram shows a field, $ABCD$, in the shape of a quadrilateral.
BD is a straight path across the field.

The total area of the field, $ABCD$, is $35\,900\,m^2$.

Work out the length of the shortest distance from D to AB.

$35900 - \frac{1}{2} \times 300 \times 112 \times \sin 140$

$35900 - 10798.03184$

$= 25101.16816 m^2$

$\frac{1}{2} \times 180 \times h = 25101.16816$

$h = \frac{25101.16816}{90}$

$h = 278.901$

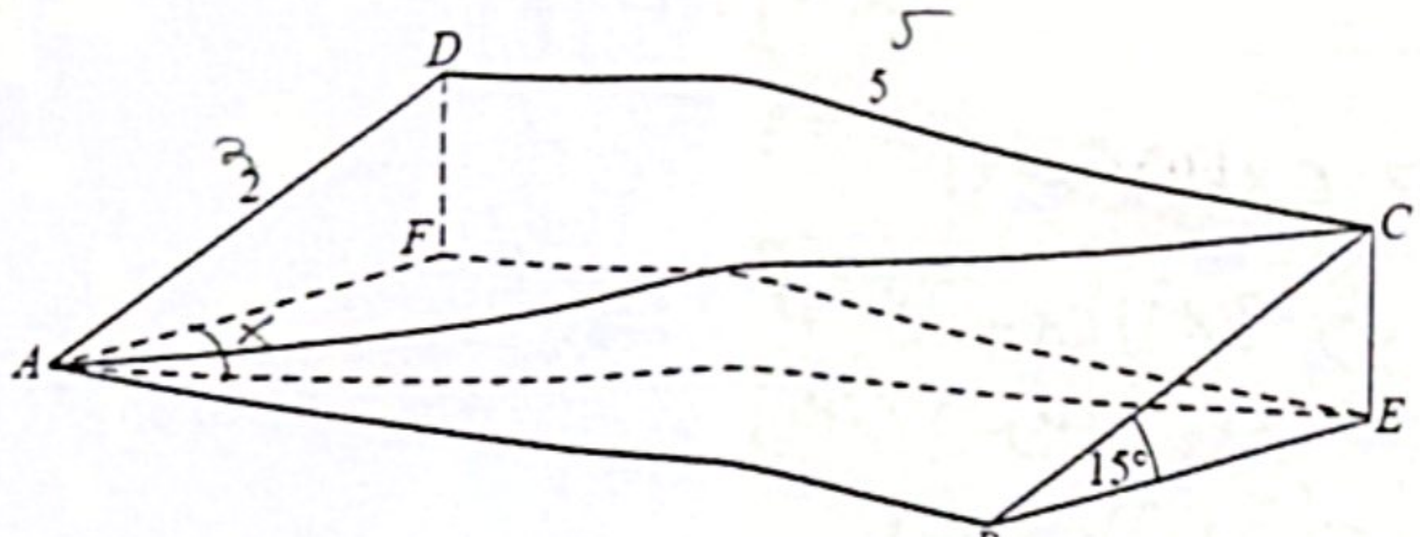
$\approx 279 m$

279 ..... m [4]

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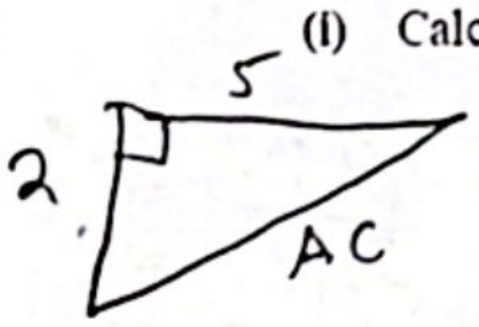
Q7.

(a)



$ABCD$ represents the rectangular sloping surface of a triangular prism.
$ABEF$ is a horizontal rectangle. CE and DF are vertical.
$C\hat{B}E = 15°$, $DC = 5$ m and $AD = 2$ m.

(i) Calculate AC.



$5^2 + 2^2 = AC^2$
$29 = AC^2$
$AC = \sqrt{29}$
$AC = 5.38516$
≈ 5.39m

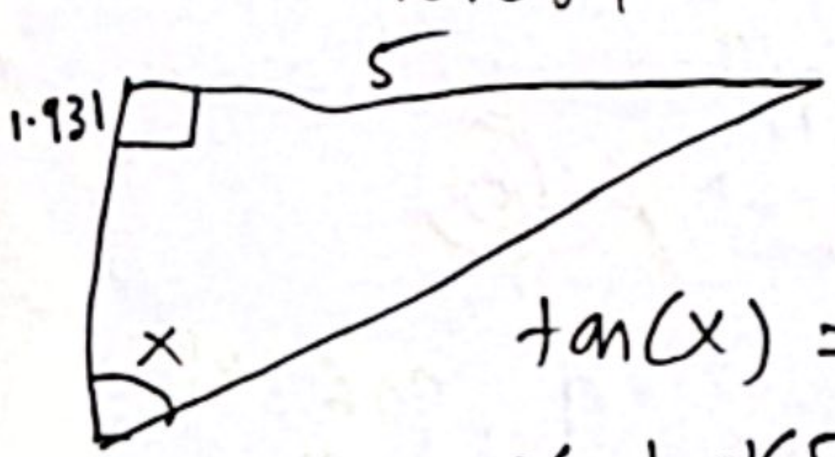
Answer 5.39 m [2]

(iii) Calculate $F\hat{A}E$.





$\cos 15 = \frac{AF}{2}$
$AF = 2\cos 15$
$AF = 1.931851$



$\tan(x) = \frac{5}{1.931}$
$x = \tan^{-1}\left(\frac{5}{1.931}\right)$
$x = 68.8749$
$\approx 68.9°$

Answer 68.9° [4]

Q8.

Simplify $\frac{x^2-16}{3x^2+10x-8}$.

$(x+4)(x-4)$

$\times 2$ $-4x+4x-16$

x^2-16 ✓

$x^2-16 \rightarrow (x+4)(x-4)$

$3x^2+10x-8 \rightarrow 3x^2+12x-2x-8$

$3x(x+4)-2(x+4)$

$\frac{(x+4)(x-4)}{(x+4)(3x-2)} \leftarrow (3x-2)(x+4)$

$\frac{x-4}{3x-2}$

$\frac{x-4}{3x-3}$ ..... [3]

Q9.

$\sin^{-1}\left(\frac{6.4}{d_2}\right) = 81.3$

$\sqrt{8.2^2 - 6.4^2} = 5.126$

$\sin^{-1}\left(\frac{5.126}{9.8}\right) = 31.54$

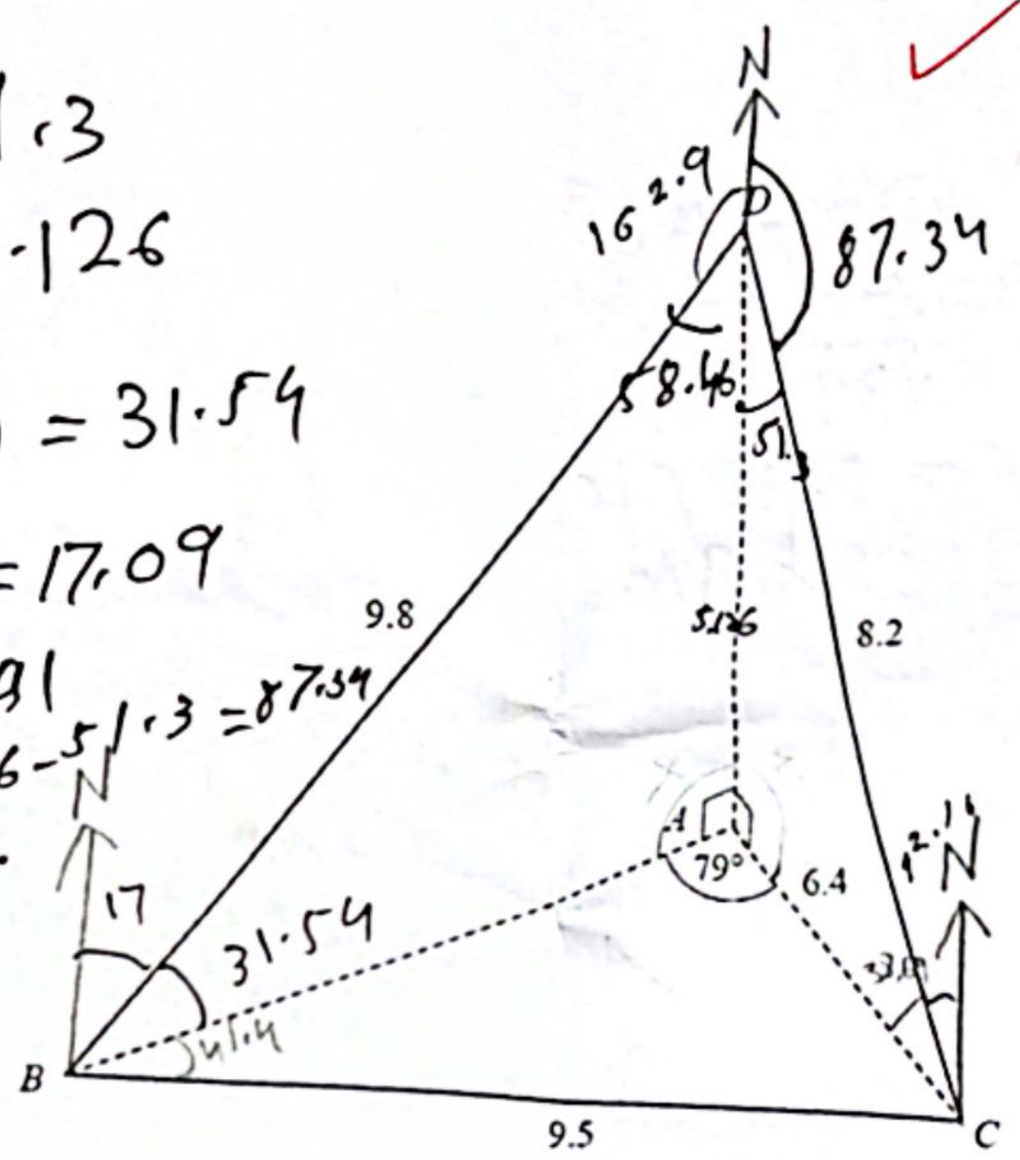
$90 - 31.54 - 41.4 = 17.09$

$180 - 17.09 = 162.91$

$360 - 162.91 - 58.46 - 51.3 = 87.34$

$180 - 87.34 = 92.66$

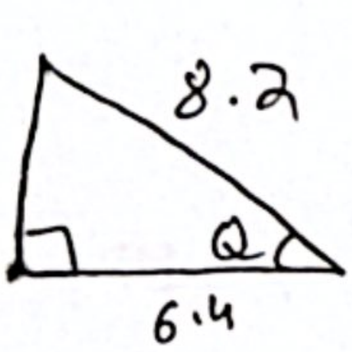
avg $= 92.66$



$\frac{\sin 31.54}{5.126} =$

The same triangle ABC forms the horizontal base of a pyramid $ABCD$.
$BD = 9.8$ cm and $CD = 8.2$ cm.
$B\hat{A}D = C\hat{A}D = 90°$

Calculate the angle of elevation of D from C.



$\cos Q = \frac{6.4}{8.2}$

$Q = \cos^{-1}\left(\frac{6.4}{8.2}\right)$

$Q = 38.694$ ✓

$\frac{\sin 79}{9.5} = \frac{\sin \hat{ABC}}{6.4}$

$\frac{6.4 \sin 79}{9.5} = \sin \hat{ABC}$

$\sin \hat{ABC} = 0.6613$

$\hat{ABC} = 41.399$

$\approx 41.4°$

$9.91°$

[2]

$\hat{ACD} = 9.906391$

≈ 9.91

$90 - 41.4 - 38.694 = \hat{ACD}$

angle of elevation.